

INITIAL BFS

	1	2	3	4	5	SUPPLY	u_i
1	16 30	16 20	13	22	17	50	
2	14	14 0	13	19 60	15	60	
3	19	19	20 10	23	14 30	50	
4	14	0	14	0	0 50	50	
DEMAND	30	20	70	30	60		
v_j							$Z = 2470 + 10M$

1. NORTH-WEST CORNER RULE

large enough to cover all the remaining demand
Eliminate column from further consideration

2. Move on the row to meet the supply ($50 - 30 = 20$)

Eliminate row from further consideration

3. Move on the column to meet the demand ($20 - 20 = 0$)

Eliminate column

4. Move on the row to meet the supply ($60 - 0 = 60$)

Eliminate row

5. Move on the column to meet the demand ($70 - 60 = 10$)

Eliminate column

6. Move on the row to meet the supply/demand ($30 < 50$)7. Move on the row to meet the supply ($50 - 10 - 30 = 10$)

Eliminate row

8. Move on the column to meet the demand ($60 - 10 = 50$)

9. Compute objective function

$$Z = 16 \cdot 30 + 16 \cdot 20 + 14 \cdot 0 + 13 \cdot 60 + 20 \cdot 10 + 23 \cdot 30 + 14 \cdot 10 + 0 \cdot 50 = 2470 + 10M$$

	1	2	3	4	5	supply	u_i
1	16 30 → 20 -2	16 20 ↓	13 -2	22 4	17 -M+22	50	-5 ^{6.}
2	14 0	14 0 → 60	13 60 ↓	18 3	15 -M+22	60	-7 ^{4.}
3	18 -2	18 -2	20 10 → 30 → 10	23 30	M 10 ↓	50	0 ^{1.}
4	M 2M-21	0 M-21	M 2M-20	0 M-23	0 5 ↓	50	-M ^{3.}
DEMAND	30	20	70	30	60	$Z = 2470 + 10M$	
V_j	21 ^{7.}	21 ^{5.}	20 ^{2.}	23 ^{2.}	M ^{2.}		

1. $u_3 = 0$ arbitrarily, the row has a lot of basic variables
 $\Rightarrow$ easier to make following computations

2. $C_{3j} - u_3 - V_j = 0 \quad j = 3, 4, 5$ because basic variables
 $\Rightarrow V_3 = 20 \quad V_4 = 23 \quad V_5 = M$

3. $C_{45} - u_4 - V_5 = 0$
 $\Rightarrow u_4 = -M$

5. $C_{22} - u_2 - V_2 = 0$
 $\Rightarrow V_2 = 21$

4. $C_{23} - u_2 - V_3 = 0$
 $\Rightarrow u_2 = -7$

6. $C_{12} - u_1 - V_2 = 0$
 $\Rightarrow u_1 = -5$

7. $C_{11} - u_1 - V_1 = 0$
 $\Rightarrow V_1 = 21$

8. All the remaining numbers can be computed with these u_i and V_j
 (each entry is $C_{ij} - u_i - V_j$)

	1	2	3	4	5	SUPPLY	u_i
1	16 30 →	16 20	13 -2	22 4	17 -M+22	50	-5
2	14 0	14 0	13 60	19 3	15 -M+22	60	-7
3	18 -2	18 -2	20 10	23 30	M 10	80	0
4	M 24-21	0 M-21	M 24-20	0 M-23	0 50	80	-M
DEMAND	30	20	20	30	60		
v_j	21	21	20	23	1000		

• not optimal because some $c_{ij} - u_i - v_j$ are negative

1. • ENTERING BASIC VARIABLE: most negative $c_{ij} - u_i - v_j \rightarrow$ to decrease the cost

=> (Source 2, Destination 5)

We mark it with +

=> INCREASED to 10 (it has to meet the demand)

2. • LEAVING BASIC VARIABLE: First that reaches 0, smallest allocation in the chain $\square$
(Source 3, Destination 5)

We mark it with -

3. • CONTINUE THE "CHAIN of REACTION" to satisfy SUPPLY and DEMAND

(S2, D5) becomes 50

(S3, D5) becomes 20

The we have to update u_i and v_j (next step)
and the cost

	1	2	3	4	5	SUPPLY	u_i
1	16 30	16 20	13 -2	22 4	17 0	50	2
2	14 0	14 0 -	13 50 +	19 3	15 10	60	0
3	19 -2	19 -2 +	20 20 -	23 30	M M - 22	50	7
4	M M + 1	0 1	M M + 2	0 -1	0 80	30	15
DEMAND	30	20	70	30	60	$Z = 2690$	
v_j	14	14	13	16	15		

NEXT STEPS (without tables, following the slides)

- ENTERING BV (S3, D2) - most negative
LEAVING BV (S2, D2) - smallest allocation
This is degenerate, no improvement in the cost
(because $C_{22} - u_2 - v_2 = 0$ hence there is a non-basic variable with a 0 in row(0))
- ENTERING BV (S1, D3) - most negative
LEAVING BV (S3, D3) - smallest allocation
- ENTERING BV (S4, D4) - degenerate
LEAVING BV (S1, D2)
- ENTERING (S3, D1)
LEAVING (S3, D4) to avoid cycling (we considered the smallest allocation already)
- ENTERING (S2, D2)
- LEAVING (S1, D1)
- DONE! $Z = 2460$